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LED array package

Abstract

Various aspects of a light emitting apparatus includes a substrate. Various aspects of the light emitting apparatus include a light emitting die arranged on the substrate. The light emitting die includes one or more side walls. Various aspects of the light emitting apparatus include a reflective die attach material extending along the one or more side walls of the light emitting die.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a division of U.S. patent application Ser. No. 18/453,164, filed on Aug. 21, 2023, which is a continuation of Ser. No. 17/735,362, filed on May 3, 2022, issued as U.S. Pat. No. 11,764,341, which is a continuation of U.S. patent application Ser. No. 17/132,325, filed on Dec. 23, 2020, issued as U.S. Pat. No. 11,355,680, which is a continuation of U.S. patent application Ser. No. 16/267,958, filed on Feb. 5, 2019, issued as U.S. Pat. No. 10,903,403, which is a continuation of U.S. patent application Ser. No. 14/449,142, filed on Jul. 31, 2014, issued as U.S. Pat. No. 10,205,069, the contents of which are incorporated herein by reference in their entireties.

BACKGROUND

Field

(1) The present disclosure relates generally to vertical LEDs and, more particularly, to vertical LEDs having light output that is similar to the light output by lateral LEDs.

Background

(2) Solid state light emitting devices, such as light emitting dies (LEDs), are attractive candidates for replacing conventional light sources such as incandescent, halogen, and fluorescent lamps. LEDs have substantially longer lifetimes than all three of these types of conventional light sources. In addition, some types of LEDs now have higher conversion efficiencies than fluorescent light sources and still higher conversion efficiencies have been demonstrated in laboratories. Finally, LEDs contain no mercury or other potentially dangerous materials, therefore, providing various safety and environmental benefits.

(3) Several different LED designs exist today, including vertical LEDs and lateral LEDs. Vertical LEDs have been considered as a replacement for lateral LEDs in array packages because vertical LEDs perform better than conventional lateral LEDs. However, simply using a vertical LED as a “drop-in” replacement for a lateral LED produces a 20% lumen drop in most LED chip-on-board configurations due to the occurrence of light absorbing effects between LEDs. Therefore, it is difficult to use vertical LEDs as a replacement for lateral LEDs without additional optimization to account for the lumen drop.

SUMMARY

(4) Several aspects of the present invention will be described more fully hereinafter with reference to various apparatuses.

(5) One aspect of a light emitting apparatus includes a substrate. The light emitting apparatus includes a light emitting die arranged on the substrate. The light emitting die includes one or more side walls. The light emitting apparatus includes a reflective die attach material extending along the one or more side walls of the light emitting die.

(6) Another aspect of the light emitting apparatus includes a substrate. The light emitting apparatus includes a light emitting die arranged on the substrate. The light emitting die includes a translucent carrier substrate.

(7) One aspect of a lamp includes a housing. The lamp includes a light emitting apparatus coupled to the housing. The light emitting apparatus includes a substrate. The light emitting apparatus includes several light emitting dies arranged on the substrate. Each of the light emitting dies includes one or more side walls. The light emitting apparatus includes a reflective die attach material extending along the one or more side walls of each of the light emitting dies.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The various aspects of the present invention illustrated in the drawings may not be drawn to scale. Rather, the dimensions of the various features may be expanded or reduced for clarity. In

addition, some of the drawings may be simplified for clarity. Thus, the drawings may not depict all of the components of a given apparatus or method.

(2) Various aspects of the present invention will be described herein with reference to drawings that are schematic illustrations of idealized configurations of the present invention. As such, variations from the shapes of the illustrations as a result, for example, manufacturing techniques and/or tolerances, are to be expected. Thus, the various aspects of the present invention presented throughout this disclosure should not be construed as limited to the particular shapes of elements (e.g., regions, layers, sections, substrates, bulb shapes, etc.) illustrated and described herein but are to include deviations in shapes that result, for example, from manufacturing. By way of example, an element illustrated or described as a rectangle may have rounded or curved features and/or a gradient concentration at its edges rather than a discrete change from one element to another. Thus, the elements illustrated in the drawings are schematic in nature and their shapes are not intended to illustrate the precise shape of an element and are not intended to limit the scope of the present invention.

(3) Various aspects of apparatuses will now be presented in the detailed description by way of example, and not by way of limitation, with reference to the accompanying drawings, wherein:

(4) FIG. **1***a* illustrates a cross-section view of an exemplary embodiment of a vertical LED.

(5) FIG. **1***b* illustrates cross-section view of a vertical LED having p-electrode routed to the top of the vertical LED.

(6) FIG. **2***a* illustrates a plan view of an exemplary embodiment of a light emitting device.

(7) FIG. **2***b* illustrates a top level view of an exemplary embodiment of a light emitting device.

(8) FIG. **3** illustrates a cross-section view of an exemplary embodiment of an LED having a die attach material.

(9) FIG. **4***a* illustrates a cross-section view of an exemplary embodiment of an LED device.

(10) FIG. **4***b* illustrates a top level view of an exemplary embodiment of the light emitting device.

(11) FIGS. **5***a*-**5***c* are side view illustrations of various exemplary apparatuses having a light-emitting device.

DETAILED DESCRIPTION

(12) The detailed description set forth below in connection with the appended drawings is intended as a description of various exemplary embodiments of the present invention and is not intended to represent the only embodiments in which the present invention may be practiced. The detailed description includes specific details for the purpose of providing a thorough understanding of the present invention. However, it will be apparent to those skilled in the art that the present invention may be practiced without these specific details. In some instances, well-known structures and components are shown in block diagram form in order to avoid obscuring the concepts of the present invention. Acronyms and other descriptive terminology may be used merely for convenience and clarity and are not intended to limit the scope of the invention.

(13) The word “exemplary” is used herein to mean serving as an example, instance, or illustration. Any embodiment described herein as “exemplary” is not necessarily to be construed as preferred or advantageous over other embodiments. Likewise, the term “embodiment” of an apparatus, method or article of manufacture does not require that all embodiments of the invention include the described components, structure, features, functionality, processes, advantages, benefits, or modes of operation.

(14) It will be understood that when an element such as a region, layer, section, substrate, or the like, is referred to as being “on” another element, it can be directly on the other element or intervening elements may also be present. In contrast, when an element is referred to as being “directly on” another element, there are no intervening elements present. It will be further understood that when an element is referred to as being “formed” on another element, it can be grown, deposited, etched, attached, connected, coupled, or otherwise prepared or fabricated on the other element or an intervening element.

(15) Furthermore, relative terms, such as “beneath” or “bottom” and “above” or “top,” may be used herein to describe one element's relationship to another element as illustrated in the drawings. It will be understood that relative terms are intended to encompass different orientations of an apparatus in addition to the orientation depicted in the drawings. By way of example, if an apparatus in the drawings is turned over, elements described as being “above” other elements would then be oriented “below” other elements and vice versa. The term “above”, can therefore, encompass both an orientation of “above” and “below,” depending of the particular orientation of the apparatus. Similarly, if an apparatus in the drawing is turned over, elements described as “below” other elements would then be oriented “above” the other elements. The terms “below” can, therefore, encompass both an orientation of above and below.
(16) It will be further understood that the terms “comprises” and/or “comprising,” when used in this specification, specify the presence of stated features, steps, operations, elements, and/or components, but do not preclude the presence or addition of one or more other features, integers, steps, operations, elements, components, and/or groups thereof. The term “and/or” includes any and all combinations of one or more of the associated listed items.
(17) Unless otherwise defined, all terms (including technical and scientific terms) used herein have the same meaning as commonly understood by a person having ordinary skill in the art to which this invention belongs. It will be further understood that terms, such as those defined in commonly used dictionaries, should be interpreted as having a meaning that is consistent with their meaning in the context of the relevant art and the present disclosure and will not be interpreted in an idealized or overly formal sense unless expressly so defined herein.
(18) In the following detailed description, various aspects of the present invention will be presented in the context of a light-emitting die. A light-emitting die shall be construed broadly to include any suitable semiconductor light source such as, by way of example, a light emitting diode (LED) or other semiconductor material which releases photons or light through the recombination of electrons and holes flowing across a p-n junction. Accordingly, any reference to an LED or other light source throughout this disclosure is intended only to illustrate the various aspects of the present invention, with the understanding that such aspects may have a wide range of applications.
(19) The following description describes a vertical LED array design for replacing lateral LED array designs. The vertical LED design described in the foregoing paragraphs provides a light output that is similar to conventional lateral LED arrays. As will be discussed in detail in the foregoing paragraphs, by providing a translucent substrate and a reflective die attach material that surrounds the vertical LEDs of an array, similar light output to conventional lateral LEDs is realized.
(20) FIG. **1***a* illustrates a cross-section view of an exemplary embodiment of a vertical LED **100**. An LED is a semiconductor material impregnated, or doped, with impurities. These impurities add “electrons” or “holes” to the semiconductor, which can move in the material relatively freely. Depending on the kind of impurity, a doped region of the semiconductor can have predominantly electrons or holes, and is referred respectively as n-type or p-type semiconductor regions.
(21) Referring to FIG. **1***a*, the LED **100** includes an n-type semiconductor region **102** and a p-type semiconductor region **105**. A reverse electric field is created at the junction between the two regions, which cause the electrons and holes to move away from the junction to form an active region **104**. When a forward voltage sufficient to overcome the reverse electric field is applied vertically across the p-n junction through a pair of electrodes **101**, **106**, electrons and holes are forced into the active region **104** and recombine. When electrons recombine with holes, they fall to lower energy levels and release energy in the form of light.
(22) In this example, the n-type semiconductor region **102** is formed on a growth substrate (not shown) and the active layer is formed between the n-type semiconductor region **102** and the p-type semiconductor region **105**. The p-type electrode **106** is directly or indirectly formed on the p-type semiconductor region **105**. The growth substrate, on which the n-type semiconductor region **102** is

formed, is removed so that the patterned n-type electrode **101** can be formed on the surface of the n-type semiconductor region **102** that was attached to the growth substrate. However, the n-type semiconductor region and the p-type semiconductor region of this example may be reversed. For instance, the p-type semiconductor region **105** may be formed on the growth substrate. As those skilled in the art will readily appreciate, the various concepts described throughout this disclosure may be extended to any suitable layered structure. Additional layers or regions (not shown) may also be included in the LED **100**, including but not limited to buffer, nucleation, contact and current spreading layers or regions, as well as light extraction layers.

(23) The n-type semiconductor layer and the p-type semiconductor layer may comprise gallium nitride. As shown in FIG. **1***a* below the n-p type semiconductor layers is a broad area reflective p-type electrode **106**, and a thermally conductive substrate **107** to support the device structure mechanically. The p-type electrode reflects light produced by the active region **104** so that it is not absorbed by the substrate **107** below. The conductive substrate may comprise a transparent or translucent material such as silicon or aluminum nitride. As will be discussed in greater detail below, aluminum nitride is preferable over silicon in chip-on-board LED packages because it absorbs less light than silicon.

(24) In the vertical LED of FIG. **1***a* the p-type electrode **106** is formed below the n-p semiconductor layers. However, in some aspects of the light emitting device, it is possible to route the p-electrode to the top of the vertical LED structure. FIG. **1***b* illustrates cross-section view of a vertical LED **120** having p-electrode **110** routed to the top of the vertical LED. As shown, both n-type electrode **101** and p-electrode **110** are above the n-p semiconductor layer. Thus, electrical traces can be connected directly to the n-type electrode **101**, providing greater flexibility in designing the vertical LED array of the chip-on-board package.

(25) FIG. **2***a* illustrates a plan view of an exemplary embodiment of a light emitting device **200**. The light emitting device includes several vertical LEDs **205** having translucent substrates **210**. As discussed in the preceding paragraphs, the translucent substrate may comprise aluminum nitride. The light emitting device also includes a substrate **220** and a phosphor layer **240**.

(26) As shown, the vertical LEDs **205** are arranged on the substrate **220**. The substrate **220** may be comprised of layers of aluminum, anodized aluminum, silver, and a distributed bragg reflector (DBR). The vertical LEDs **205** are also covered by the phosphor layer **240** which scatters some of the light produced by the vertical LEDs **205**. The phosphor layer, which will be discussed in greater detail with respect to FIG. **2***b*, may comprise phosphor particles that scatter the light from the vertical LEDs in different directions. The scattered light, in some aspects of the device, could be redirected to an adjacent vertical LED. If the adjacent vertical LED comprises material that is light absorbing, at least some of the redirected light will be absorbed by the adjacent vertical LED. When light is absorbed rather than output, the device's performance suffers.

(27) The vertical LEDs **205** in this exemplary figure have a translucent substrate **210**, which has lower light absorption properties than commonly used substrate materials such as silicon. Thus, light absorption between adjacent vertical LEDs is reduced by the translucent substrate **210** leading to better light output from the light emitting device. For instance, the translucent substrate may provide a 10% lumen increase in light output over a conventional silicon substrate. Additionally, the substrate may be optically tuned to make the substrate more or less translucent so that the LED array can be further customized for different devices.

(28) FIG. **2***b* illustrates a top level view of an exemplary embodiment of the light emitting device **200**. The light emitting device includes an array of the vertical LEDs **205**, the phosphor layer **240**, and the substrate **220**. As shown, the substrate **220** may be used to support a number of vertical LEDs **205**. The phosphor layer **240** may be deposited within a cavity defined by an annular, or other shaped boundary that extends around the upper surface of the substrate **220**. The boundary may be formed by a suitable mold or, alternatively, formed separately from the substrate **220** and attached to the substrate using an adhesive or other suitable means. The phosphor layer **240** may

include phosphor particles suspended in an epoxy, silicone, or other carrier that may be constructed from soluble phosphor that is dissolved in the carrier. As discussed above with respect to FIG. **2***a*, light emitted from each of the vertical LEDs **205** may be scattered by the phosphor layer and partially absorbed by adjacent LEDs. However, the translucent substrates **210** of the vertical LEDs **205** partially remedies the absorption issue because it absorbs less light than other conventional substrates such as silicon. However, the incorporation of a reflective adhesive material that surrounds the vertical LED may further remedy light absorption issues between vertical LEDs in a chip-on-board package.

(29) FIG. **3** illustrates a cross-section view of an exemplary embodiment of a vertical LED **300** attached to a substrate **320** by a die attach material **305**. The vertical LED **300** includes a translucent substrate **310** as described in detail with respect to FIGS. **1***a*-**1***b* and **2***a*-**2***b*, and is arranged on the substrate **320**. The die attach material **305** is used to bond the vertical LED **300** to the substrate **320**. The die attach material also covers at least portion of the side walls **325** of the vertical LED **300**. In some aspects of the device, the die attach material may cover a portion of the side wall **325** of the vertical LED **300** that at least half the height of the vertical LED. In such aspects, the die attach material **305** may cover a portion of the vertical LED **300** that is up to, but not over the active region **104**, which was discussed in detail with respect to FIGS. **1***a*-**1***b*.

(30) The die attach material increases the light output of vertical LEDs arranged in an array by decreasing light absorption between the vertical LEDs that are arranged close to each other. In some aspects of the device, the die attach material may be a white epoxy because white epoxy forms a reflective surface. However, the die attach material may be any adhesive suitable for bonding the vertical LED to the substrate **320** that also has reflective properties. By limiting light absorption further, the reflective die attach material may boost the light output of a vertical LED array by another 10% lumen (in addition to the 10% lumen increase produced by the translucent substrate).

(31) FIG. **4***a* illustrates a cross-section view of an exemplary embodiment of light emitting device **400**. The light emitting device **400** includes several vertical LEDs **300** arranged on a substrate **420**. The LED device **400** of FIG. **4***a* is similar to the LED device **200** of FIGS. **2***a*-**2***b* in that both devices include several LEDs arranged on a substrate. However, as will be discussed, LED device **400** is designed to perform better than the light emitting device **200** because to the inclusion of die attach material **305** provides a higher light output.

(32) Referring to FIG. **4***a*, as shown, the die attach material **305** is used to bond each of the LEDs **300** to the substrate **420**. The substrate **420** may include material similar to that of substrate **220** described with respect to FIG. **2***a*. Additionally, the die attach material covers at least a portion of the side walls of each of the vertical LEDs **300**.

(33) As discussed above, the die attach material **305** comprises a reflective material such as white epoxy. As a result, light that is scattered by the phosphor layer is reflected by the die attach material that would have otherwise been absorbed by the vertical LED.

(34) As illustrated in FIG. **4***a*, the die attach material increases light output of the LED device **400** because light absorption between LEDs is reduced. Thus, more light is reflected away from adjacent vertical LEDs by the die attach material **305**. In some aspects of the device, this design increases light output by 10% lumen. Thus, the combination of a translucent substrate and die attach material produces a light output similar to the light output of conventional lateral LEDs, making this design a suitable replacement for lateral LEDs in a chip-on-board configuration.

(35) FIG. **4***b* illustrates a top level view of an exemplary embodiment of the light emitting device **400**. The light emitting device **400** includes similar features to those of light emitting device **200** described with respect to FIG. **2***b*. For instance, the light emitting device **400** includes an array of the vertical LEDs **300**, the phosphor layer **440**, and the substrate **420**. However, the light emitting device **400** additionally includes the reflective die attach material **305** that surrounds the side walls of each of the LEDs **300** in the array. Thus, greater light output is realized by the light emitting

device because the die attach material prevents light that is scattered by phosphor layer **440** from being absorbed by the vertical LEDs **300**.
(36) FIG. **5***a* is a side view illustration of an exemplary lamp **500** having a light emitting device **502**. Lamp **500** may be used for any type of general illumination. For example, lamp **500** may be used in an automobile headlamp, street light, overhead light, or in any other general illumination application. The light emitting device **502** may be located in a housing **506**. The light emitting device **502** may receive power via a power connection **504**. The light emitting device **502** may be configured to emit light. Description pertaining to the process by which light is emitted by the light-emitting device **202** is provided with reference to FIGS. **4***a*-**4***b*.
(37) FIG. **5***b* is a side view illustration of a flashlight **510**, which is an exemplary embodiment of an apparatus having the light emitting device **502**. The light emitting device **502** may be located inside of the housing **506**. The flashlight **510** may include a power source. In some aspects of the light emitting device, the power source may include batteries **514** located inside of a battery enclosure **512**. In another aspect of the light emitting device, power source **910** may be any other suitable type of power source, such as a solar cell. The power connection **504** may transfer power from the power source (e.g., the batteries **514**) to the light-emitting device **502**.
(38) FIG. **5***c* is a side view illustration of a street light **520**, which is another exemplary embodiment of an apparatus having the light emitting device **502**. The light emitting device **502** may be located inside of the housing **506**. The street light **520** may include a power source. In some exemplary embodiments, the power source may include a power generator **522**. The power connection **504** may transfer power from the power source (e.g., the power generator **522**) to the light emitting device **502**.
(39) The various aspects of this disclosure are provided to enable one of ordinary skill in the art to practice the present invention. Various modifications to exemplary embodiments presented throughout this disclosure will be readily apparent to those skilled in the art, and the concepts disclosed herein may be extended to other devices. Thus, the claims are not intended to be limited to the various aspects of this disclosure, but are to be accorded the full scope consistent with the language of the claims. All structural and functional equivalents to the various components of the exemplary embodiments described throughout this disclosure that are known or later come to be known to those of ordinary skill in the art are expressly incorporated herein by reference and are intended to be encompassed by the claims. Moreover, nothing disclosed herein is intended to be dedicated to the public regardless of whether such disclosure is explicitly recited in the claims. No claim element is to be construed under the provisions of 35 U.S.C. § 112(f) unless the element is expressly recited using the phrase "means for" or, in the case of a method claim, the element is recited using the phrase "step for."

Claims

1. A light emitting apparatus comprising: a base substrate; a light emitting die coupled to the base substrate and having an active region configured to emit light, wherein the light emitting die comprises a vertical light emitting diode (LED); and a p-electrode formed on a bottom of a p-type semiconductor region and on a top of a surface of a n-type semiconductor region such that a portion of the p-electrode is routed to a top of the vertical LED and a n-type electrode formed on the top of the surface of the n-type semiconductor region disposed on the light emitting die opposite to the base substrate, wherein the active region is formed between the n-type semiconductor region and the p-type semiconductor region.
2. The light emitting apparatus of claim 1, wherein the light emitting die comprises one or more side walls, wherein at least a portion of an outer side wall of the light emitting die is covered with a reflective die attach material that reduces light absorption.
3. The light emitting apparatus of claim 2, wherein the die attach material extends along at least

half of a height of the one or more side walls of the light emitting die.
4. The light emitting apparatus of claim 2, wherein the light emitting die further comprises a transparent substrate disposed between the one or more side walls.
5. The light emitting apparatus of claim 2, wherein the die attach material comprises white epoxy to form a reflective surface for limiting light absorption.
6. The light emitting apparatus of claim 1, wherein the light emitting die comprises a gallium nitride layer over an aluminum nitride substrate.
7. The light emitting apparatus of claim 6, further comprising a reflective layer between the gallium nitride layer and the aluminum nitride substrate.
8. The light emitting apparatus of claim 1, further comprising a plurality of light emitting dies arranged on the substrate.
9. The light emitting apparatus of claim 2, wherein the reflective die attach material extends in a vertical direction and covers at least a portion of an outer side of one or more side walls of the substrate of the light emitting die.
10. The light emitting apparatus of claim 9, further comprising a phosphor layer covering a plurality of light emitting dies, wherein the phosphor layer is configured to emit scattered light.
11. The light emitting apparatus of claim 10, wherein the reflective die attach material is directly secured to each of a plurality of light emitting dies.
12. A light emitting apparatus comprising: a substrate; a die attach material; a light emitting die coupled to the substrate with the die attach material and having a carrier substrate translucent and an active region configured to emit light, wherein the light emitting die comprises a vertical light emitting diode (LED); and a p-electrode formed both on a bottom of a p-type semiconductor region and on a top of a surface of a n-type semiconductor region such that a portion of the p-electrode is routed to a top of the vertical LED and a n-type electrode formed on the top of the surface of the n-type semiconductor region, wherein the active region is formed between the n-type semiconductor region and the p-type semiconductor region.
13. The light emitting apparatus of claim 12, wherein the light emitting die further comprises one or more side walls, wherein at least a portion of an outer side wall of the light emitting die is covered with a reflective die attach material that reduces light absorption.
14. The light emitting apparatus of claim 12, wherein the carrier substrate comprises aluminum nitride.
15. The light emitting apparatus of claim 12, further comprising a reflective layer arranged over the carrier substrate.
16. The light emitting apparatus of claim 12, further comprising a plurality of light emitting dies arranged on the substrate.
17. The light emitting apparatus of claim 16, further comprising a phosphor layer arranged over the plurality of light emitting dies, wherein the phosphor layer is configured to emit scattered light.
18. The light emitting apparatus of claim 13, wherein the reflective die attach material extends in a vertical direction and covers at least a portion of an outer side of one or more side walls of the substrate of the light emitting.
19. A lamp comprising: a housing; and a light emitting apparatus coupled to the housing, the light emitting apparatus comprising: a substrate; a light emitting die coupled to the substrate and having an active region configured to emit light, wherein the light emitting die comprises a vertical light emitting diode (LED); and a p-electrode formed on both a bottom of a p-type semiconductor region and on a top of a surface of a n-type semiconductor region such that a portion of the p-electrode is routed to a top of the vertical LED and a n-type electrode formed on the top of a surface of the n-type semiconductor region, wherein the active region is formed between the n-type semiconductor region and the p-type semiconductor region.